Atreya Vedantam

ee22b004@smail.iitm.ac.in · atreyavedantam.github.io · Google Scholar · github.com/atreyavedantam

Theory of Deep Learning · Optimization · Feature Learning · Generalization · Privacy · Statistics

Education

Indian Institute of Technology Madras Nov 2022 – Jul 2027 (expected)

Chennai, India

Dual Degree: B.Tech. and M.Tech. in Electrical Engineering

CGPA **9.65**/10 Department rank **2** of 175

Master's thesis: *Memory-sample tradeoffs in high-dimensional statistics.* Advisors: Vatsal Sharan (USC), Kabir Verchand (USC), Ankit Pensia (CMU).

Bachelor's thesis: *Optimizing Entanglement Distillation and Swap Decisions in Homogeneous Quantum Repeater Chains.* Advisor: Krishna Jagannathan (IIT Madras).

Publications and Preprints

* denotes equal contribution.

Conference papers

1. **A. Vedantam**, R. K. Ganti. *Characterization of Blind Code Rate Recovery in Linear Block Codes.* IEEE International Symposium on Information Theory (**ISIT**), 2026. *Presented in person.*

Preprints and papers under review

2. **A. Vedantam***, N. Das*, C. Lakshminarayanan. *On the Dynamics of Delayed Generalization in Linear Models*. arXiv preprint 2602.08302, 2026. Under review at **AISTATS** 2027.
3. **A. Vedantam**, K. Jagannathan. *Optimizing Entanglement Distillation and Swap Decisions in Homogeneous Quantum Repeater Chains.* Under review at **SIGMETRICS** 2027.

Research Experience

Feature Learning under Spectral Optimizers May 2026 – present

University of Southern California · Prof. Vatsal Sharan · IUSSTF-Viterbi Program

- Generalized the Neural Feature Ansatz (NFA) to account for the optimizer. Showed that under spectral optimizers (Muon, spectral GD) the standard NFA fails, while a spectrum-aware form tracks the learned features.
- Derived the average gradient outer product (AGOP) as an optimal low-rank compression of the input; proposed a covariance-weighted AGOP for non-isotropic data. Tested the predictions from MLPs up to GPT-2, using subspace-overlap and spectral diagnostics.

Memory-Sample Tradeoffs for Testing May 2026 – present

USC and CMU · Profs. Vatsal Sharan, Kabir Verchand, Ankit Pensia · Master's thesis

- Proved memory-sample lower bounds for the testing version of planted-matrix streaming problems, building on Raz (2017) and Garg–Raz–Tal (2018).
- The general theorem recovers and strengthens earlier results of Garg, Kothari and Raz as a corollary.

Optimal Membership Inference Attacks on Linear Models Jan 2026 – present

IIT Madras · Prof. Krishna Pillutla

- Characterized optimal membership inference attacks on ridge regression models and derived per-point susceptibility certificates in terms of the residual, ridge leverage and effective dimension.
- Working on the optimal privacy preserving mechanism for fine tuning a ridge regression model.

Delayed Generalization in Linear Models Oct 2025 – Aug 2026

Machine Learning and Learning Theory Lab, IIT Madras · Prof. Chandrashekar Lakshminarayanan

- Obtained non-asymptotic results of delayed generalization in linear classifiers trained by gradient descent: an imbalance in the support vectors drives the bias through three distinct phases.

Entanglement Distribution in Quantum Networks Aug 2025 – Oct 2026

IIT Madras · Prof. Krishna Jagannathan · Bachelor's thesis

- Modeled entanglement distribution in homogeneous quantum repeater chains and characterized when distillation helps; derived an analytically tractable distillation model and optimized swap decisions.

Blind Code Rate Recovery in Linear Block Codes Jan 2025 – Jan 2026
SATLAB, IIT Madras · Prof. Radha Krishna Ganti

- Showed that no rank-based method can recover the code rate blindly much beyond a signal-to-noise ratio threshold, and derived optimal recovery parameters at a given SNR. Presented at ISIT 2026.

Real-time Spectrograms with Dispersive Fibres May 2024 – Nov 2024
IIT Madras · Profs. Deepa Venkitesh and L. N. Theagarajan

- Derived from first principles a closed-form expression for the time-mapped short-time Fourier transform computed by a dispersive fibre, and studied optimal fibre parameters for real-time spectrogram computation.

Quantum Photon Detectors with Metasurfaces Oct 2023 – 2024
IIT Madras · Prof. Praveen Bhallamudi

- Designed metaphotonic surfaces for single-photon detection at 1550 nm, improving absorptivity and spectral selectivity; simulated in ANSYS Lumerical.

Industry Experience

Quantitative Research Intern May 2025 – Jul 2025
Dynamic Technology Lab, Singapore

- Built statistical and machine-learning models for alpha generation from mutual-fund position data; designed a spectrogram-based co-integration metric for stock pairs.

Teaching Experience

Teaching Assistant, IIT Madras

- Convex Optimization
- Multirate Digital Signal Processing
- Introduction to Wireless and Cellular Communication
- Signals and Systems
- Engineering Electromagnetics

Honors and Awards

IUSSTF-Viterbi Research Scholar – funded research fellowship at USC; about 15 selected from 400+ 2026
Department Rank 2 – Electrical Engineering, IIT Madras, among 175 students 2026
KVPY Fellowship (SX) – All India Rank 475 among about 50,000 candidates 2021
JEE Advanced – All India Rank 315 2022
JEE Main – All India Rank 221; 99.98 percentile among 1,026,799 candidates 2022
Inter-IIT AI Hackathon (Convolve) – top 5 in the final round among 1500+ teams

Leadership and Service

Founder, Srinivasa Ramanujan Mathematics Competition (SRMC) 2024 – 2025
IIT Madras

- Founded India's first olympiad-style mathematics competition for engineering undergraduates, modelled on the Putnam.
- Launched without sponsorship by contacting 100 colleges directly: 700 registrations in year one, sponsorship from Jane Street in year two, and 2,200+ participants in year three.

Head, Mathematics Club 2024 – 2025
IIT Madras · previously Coordinator and Deputy Coordinator

- Led a team of 70+. Ran the club's Summer School (about 1000 registrations nationwide) and launched its first website.
- Started the Integration Bee at IIT Madras (150+ participants) and ran the Combino-Geo Grand Prix (200+ participants).
- Led a project on the Travelling Salesman Problem that produced a heuristic orders of magnitude faster than simulated annealing.

Coordinator, AI Club 2024 – 2025
Centre for Innovation, IIT Madras

- Designed and ran the Shaastra Techathon AI/ML Challenge 2024 for 350+ participants from 20+ colleges: problem statement, dataset, evaluation metric and introductory session.

Mentorship 2022 – 24

IIT Madras

- **Avanti Fellows** (2022 – 23): Mentored two students from under-resourced schools for the JEE; both joined IIT Madras.
- **SATHEE** (2023): recorded JEE problem-solving and concept videos for a national free-preparation platform; ran online doubt sessions.
- **Saathi** (2023–24): mentored four first-year students through their first year at IIT Madras.

Selected Graduate Coursework

Convex Optimization · Foundations of Machine Learning · Fundamentals of Deep Learning · Privacy in AI · Information Theory · Computability and Complexity · Stochastic Modeling and the Theory of Queues · Quantum Computation and Quantum Information · Multirate Digital Signal Processing

Beyond Research

- **Author:** published a 250-page fantasy novel, *The Tragedy of Girion*, at the age of 13 (2018), five years after beginning it.
- **Music:** trained in Carnatic (Indian classical) vocal music for about ten years.
- **Sport:** selected for the National Sports Organisation in tennis and in badminton; probable for the IIT Madras men's tennis contingent.